

AYA ARROCHE

Emerging Technologies Engineer | XR Developer | AI-Driven Interactive Systems

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PROFESSIONAL SUMMARY

Emerging Technologies Engineer (Master's, UCA – ENS Marrakech) specialized in XR and AI-driven interactive systems, with a solid software engineering foundation across backend architecture, REST/microservices, and DevOps pipelines. Experienced in building immersive, adaptive experiences — from real-time Unity/XR gameplay systems to AI pipelines and Dockerized full-stack deployments. Strong ability to turn ideas into structured technical documentation and to collaborate across engineering, design, and AI disciplines. Adaptable, fast learner, motivated to grow within a large-scale engineering environment.

TECHNICAL SKILLS

- XR & Emerging Tech:** Unity XR/VR development, MediaPipe (computer vision), real-time sensor-driven interaction, immersive simulation design
- Game Dev & AI:** Unity (C#), Gameplay Programming, WebGL, Android (Java), LangChain, LLM integration, Machine Learning
- Backend & APIs:** FastAPI, Flask, Spring Boot, REST API, WebSocket, Microservices, RabbitMQ
- DevOps & Cloud:** Docker, Docker Compose, Jenkins, GitHub Actions, SonarQube, CI/CD
- Databases & Tools:** PostgreSQL, MySQL, Git, GitHub, VS Code, Android Studio

EXPERIENCE

- Game Developer — Internship** — Kokoro Games, Casablanca (Remote) Jul. – Aug. 2026 · 2 months
- Brought strong initiative and a clear product vision to the studio's game pipeline, structuring and documenting ideas into comprehensive technical and design documentation while collaborating closely with cross-functional teams across multiple game projects
- IT Teacher** — High School Al Massar, Marrakech Mar. – Jul. 2024 · 4 months
- Simplified the IT curriculum for college-level students and introduced new pedagogical methods to improve learning outcomes
- IT Support Technician** — Collège Elia Abou Madi, Marrakech Apr. – Jun. 2023 · 3 months
- Digitized the institution's paper-based administrative records into a desktop system, streamlining day-to-day office operations

KEY PROJECTS

- [PFE] EduForge — Adaptive Serious Game** 2025 – 2026
- Unity (C#) · Gameplay Programming · MediaPipe · FastAPI · PostgreSQL · WebSocket · LangChain · LLaMA-3.1 · xAPI
- Designed and programmed a complete Unity game with real-time gameplay logic that adapts difficulty live based on the player's emotional state, using a custom Dynamic Difficulty Adjustment system grounded in Flow theory
 - Integrated real-time facial emotion detection (MediaPipe) directly into the gameplay loop via a WebSocket bridge to a FastAPI/PostgreSQL backend
 - Connected an LLM pipeline (LangChain + Groq/LLaMA-3.1) to generate dynamic in-game content, plus full xAPI analytics; deployed end-to-end on a live VPS with Docker
- Microservices Architecture & DevOps Pipeline** 2026
- Java · Spring Boot · Docker · Jenkins · RabbitMQ · MySQL · GitHub Actions · SonarQube
- Architected a microservices system with REST APIs, asynchronous RabbitMQ messaging, Docker containerization, and a full CI/CD pipeline
- Urban Simulation & AI Platform** 2025
- Python · Flask · Machine Learning · scikit-learn · Data Visualization
- Built an ML-powered urban simulation with a Flask RESTful API, real-time environmental data ingestion, and monitoring dashboards
- Android Applications — PizzaRecipes & Geolocation App** 2025
- Java · Android Studio · Google Maps SDK · DAO/MVC Architecture · REST API · GPS
- Built two Android apps: a recipe app with clean DAO/Service/Adapter architecture, and a GPS/Google-Maps app consuming a REST API in real time

EDUCATION

- Master's — Engineering and Emerging Technologies · UCA – ENS Marrakech · 2024 – 2026**
- Bachelor's — Computer Science · UCA – ENS Marrakech · 2021 – 2024**

CERTIFICATIONS & LANGUAGES

- Certifications:** NLP in Python · Front-End Dev with React · React Native · Java Advanced · Hibernate & JPA
- Languages:** Arabic (Native) · French (B2) · English (C1)